Table 8
RoBERTa-Large is fine-tuned on examples with different depths from PARARULES and also the entire PARARULE-Plus(PPT), and then is evaluated on test sets that require different depths of reasoning. The yellow background indicates improvement on accuracy after adding our PARARULE-Plus in the training process.

Model	Test set	Mod012 (Depth≤2+PPT)	Mod0123 (Depth≤3+PPT)	Mod0123Nat (Depth≤3+NatLang+PPT)	Mod012345 (Depth≤5+PPT)
RoBERTa-Large	Depth=0	0.946	0.901	0.965	**0.963 (+0.010)**
	Depth=1	0.877	0.847	**0.937 (+0.004)**	0.881
	Depth=2	0.868	0.873	**0.927**	0.839
	Depth=3	0.771 (+0.209)	0.862	**0.904**	0.826
	Depth=4	0.675 (+0.194)	0.852	**0.897**	0.832
	Depth=5	0.661 (+0.209)	0.888 (+0.032)	0.923 (+0.007)	**0.934 (+0.001)**
	NatLang	0.557	0.593 (+0.014)	**0.970 (+0.008)**	0.649 (+0.055)

imbalance issue of the current datasets and the addition of it during training improves the model's generalisation on the examples that require deeper reasoning steps.

7. Conclusion

We provide insights into an RNN-based iterative memory model that incorporates gated attention on multi-step reasoning over natural language. Instead of using the original GRU and dot-product attention, we integrate gated attention to update hidden states. The experiment results show the model with gated attention achieves generally better performance than the original RNN-based iterative-memory model with dot-product attention and other RNN-based models. The performance of our model is comparable or better than the much larger and pretrained RoBERTa-Large in some scenarios. Furthermore, our model shows better out-of-distribution generalisation performance than the pretained RoBERTa. To address the issue of depth-imbalance in the existing datasets on multi-step reasoning over natural language, we develop a large-scale multi-step reasoning dataset called PARARULE-Plus, with more examples of deep reasoning depths than previous datasets. We find that the performance of the models in our experiments improves when we add PARARULE-Plus in the training, especially on examples that require deeper reasoning depths and extra out-of-distribution examples.

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